

Experiment No.	5
Experiment Title	Decision Tree and Random Forest - A Comparative Classification Study
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Submission Date	17/08/2026

1 Objective

The idea of this experiment is to build a Decision Tree classifier first, then push it further into a Random Forest ensemble and see how much of a difference the ensembling actually makes. Concretely, the goals are to:

- implement a Decision Tree classifier from the scikit-learn library,
- extend it into a Random Forest ensemble,
- study how hyperparameters like tree depth control overfitting versus generalization,
- pick the best hyperparameters using 5-fold cross-validation instead of eyeballing a single split, and
- compare the tuned single tree against the tuned forest on a held-out test set.

2 Problem Statement

Given a set of numerical features computed from digitized images of breast masses (cell nuclei measurements such as radius, texture, perimeter, area, smoothness, etc.), the task is to classify each sample as malignant or benign. This is a binary classification problem where the input is a 30-dimensional feature vector per sample and the output is a diagnosis label. Since this is a medical diagnosis setting, the objective isn't just raw accuracy recall on the malignant class matters a lot too, because a missed malignant case is a far more costly mistake than a false alarm on a benign one.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository, loaded via <code>sklearn.datasets.load_breast_cancer</code>
Number of Samples	569
Number of Features	30 numerical features (10 base measurements × mean, standard error, and worst value)
Number of Classes	2 Malignant (M) and Benign (B)
Missing Values	None found in the loaded dataset

Train-Test Split	80% train (455 samples) / 20% test (114 samples), stratified on the class label
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4 Exploratory Data Analysis

Class Distribution

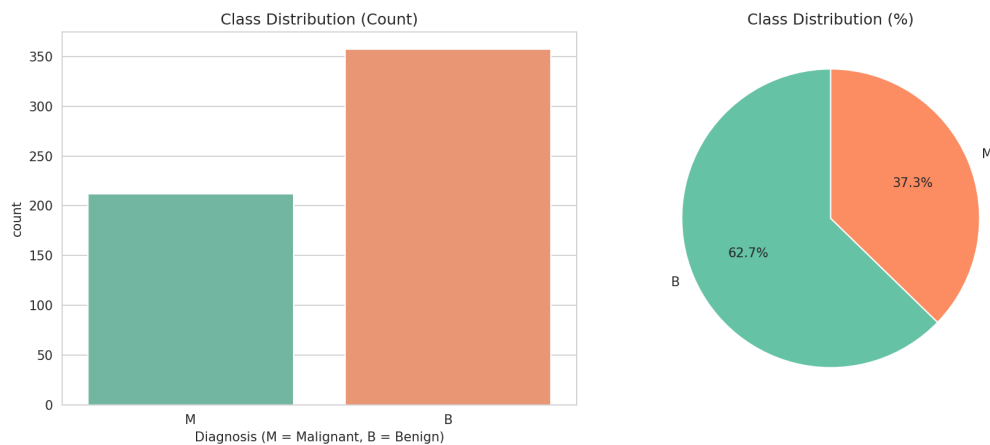


Figure 1: Class distribution of the WDBC dataset - counts (left) and proportion (right).

Inference:

The dataset is moderately imbalanced: 357 samples (62.7%) are benign and 212 (37.3%) are malignant. It isn't a severe imbalance, so accuracy is still a reasonable headline metric here, but it's skewed enough that precision/recall and a stratified split matter - which is exactly why the train-test split and the cross-validation folds were both stratified. A model that just predicted "benign" every time would already be right almost two-thirds of the time, which is a useful sanity-check baseline to keep in mind against the actual results below.

Feature Correlation Analysis

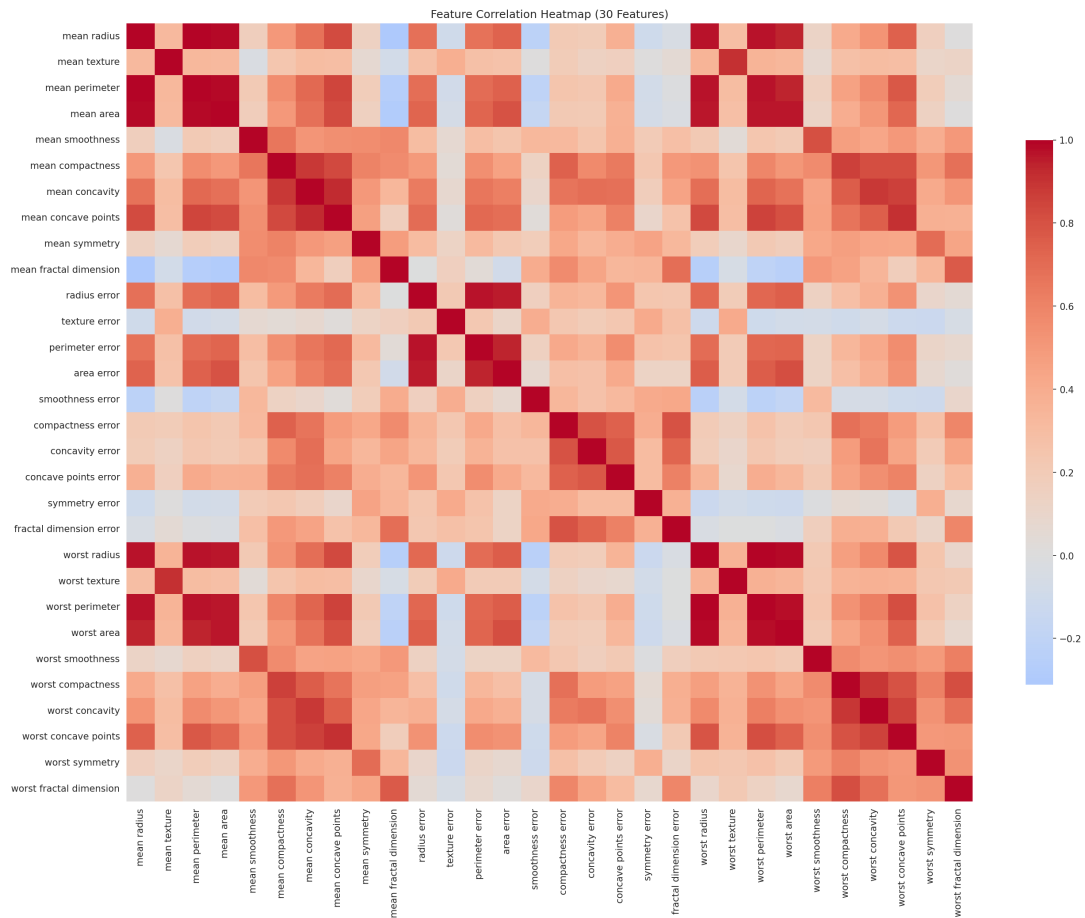


Figure 2: Correlation heatmap across all 30 features.

Inference:

The heatmap makes the redundancy in this dataset pretty obvious - **mean radius**, **mean perimeter** & **mean area** are almost perfectly correlated with each other (which makes sense geometrically, they are all describing the size of the same cell nucleus), and the same pattern repeats for their **worst** counterparts. The error features stand mostly on their own and don't correlate strongly with much else. This kind of multicollinearity doesn't hurt tree-based models the way it would hurt something like logistic regression, since a tree just picks whichever correlated feature gives the best split at each node, but it's still useful context for interpreting the feature-importance plot later on.

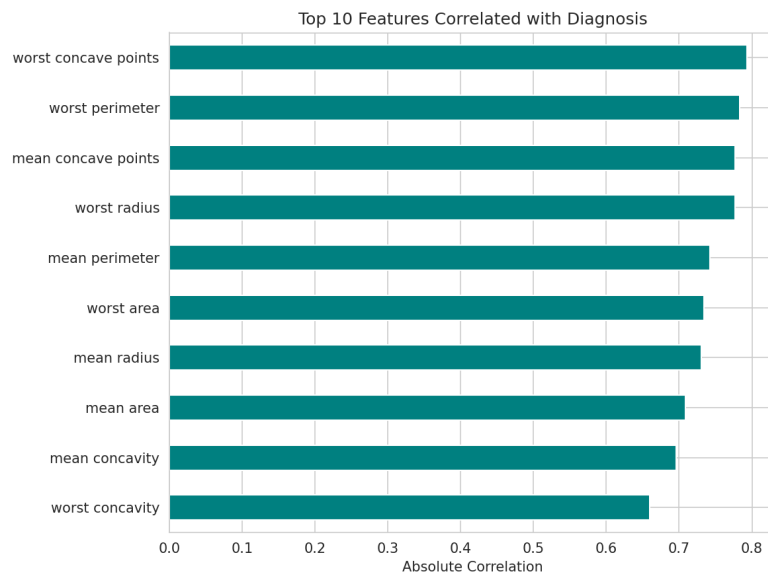


Figure 3: Top 10 features by absolute correlation with the diagnosis label.

Inference:

worst concave points, worst perimeter, mean concave points and worst radius come out on top, all with correlations above 0.77 with the target. Broadly, features describing size (radius, perimeter, area) and shape irregularity (concavity, concave points) separate the two classes well on their own - larger, more irregularly-shaped nuclei tend to be malignant. This lines up with the medical intuition behind the dataset and, as it turns out later, lines up closely with what the Random Forest actually ends up using most.

5 Data Preprocessing

The dataset needed very little cleaning, which is one of the nicer things about using the built-in sklearn version of WDBC:

- **Missing values:** none present, so no imputation was needed.
- **Encoding:** the target was already numerically encoded by sklearn (0 = malignant, 1 = benign), but for clarity a human-readable diagnosis column ('M'/'B') was added and then re-encoded explicitly using `LabelEncoder` as the lab steps require - this maps $B \rightarrow 0$ and $M \rightarrow 1$ alphabetically.
- **Feature scaling:** not applied. Decision Trees and Random Forests split on threshold comparisons of individual features, so they're invariant to monotonic rescaling - scaling wouldn't change the tree structure at all, so it was skipped rather than added for no benefit.
- **Train-test split:** an 80-20 stratified split (`random_state=42`) was used so that the roughly 63/37 class balance is preserved in both the training and test sets.
- **Feature engineering:** none - all 30 original features were retained and fed directly into both models.

6 Machine Learning Algorithm

Decision Tree

A Decision Tree recursively splits the feature space into regions by repeatedly choosing the feature and threshold that best separates the classes at each node, typically using an impurity measure such as Gini impurity or entropy/information gain. The splitting continues until a stopping criterion is met (maximum depth, minimum samples per leaf, or pure leaves). **Advantages:** easy to interpret and visualize, requires no feature scaling, and can capture non-linear decision boundaries. **Limitations:** prone to overfitting when grown deep, and small changes in the training data can produce a noticeably different tree (high variance).

Random Forest

A Random Forest builds an ensemble of many Decision Trees, each trained on a bootstrap sampled subset of the training data (bagging), and at every split only a random subset of features is considered (`max_features`). Final predictions are made by majority vote across all trees. **Advantages:** averaging over many de-correlated trees reduces variance and generally improves generalization compared to a single tree, and it naturally provides a feature-importance ranking. **Limitations:** loses much of the easy interpretability of a single tree, is more expensive to train and to run inference with, and can still overfit if individual trees are allowed to grow very deep with too little diversity between them.

7 Experimental Setup

Python Version	3.12
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Linux
Hardware	CPU only

8 Hyperparameter Selection

Both models were tuned with GridSearchCV over 5-fold stratified cross-validation on the training set.

Decision Tree - Search Space and Best Configuration

Grid: `criterion` $\in$ {gini, entropy}, `max_depth` $\in$ {3, 5, 7, 10, None}, `min_samples_split` $\in$ {2, 5, 10}, `min_samples_leaf` $\in$ {1, 2, 4}.

Parameter	Value
<code>criterion</code>	entropy
<code>max_depth</code>	5
<code>min_samples_split</code>	10
<code>min_samples_leaf</code>	2

Best 5-fold CV accuracy for this configuration: 93.63%.

Random Forest - Search Space and Best Configuration

Grid: `n_estimators` $\in \{50, 100, 200\}$, `max_depth` $\in \{5, 10, \text{None}\}$, `max_features` $\in \{\text{sqrt}, \text{log2}\}$, `bootstrap` $\in \{\text{True}, \text{False}\}$.

Parameter	Value
<code>n_estimators</code>	200
<code>max_depth</code>	5
<code>max_features</code>	sqrt
<code>bootstrap</code>	False

Best 5-fold CV accuracy for this configuration: 97.14% - already a good hint that the forest is going to come out ahead of the single tree.

9 Results

Test Set Performance

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	0.9649	1.0000	0.9048	0.9500
Random Forest	0.9649	1.0000	0.9048	0.9500

Interestingly, the two models land on identical point metrics on this particular 114-sample test set (both misclassify the same 4 malignant cases as benign, and get every benign case right). Where they actually separate is the ROC-AUC, discussed below - the Random Forest's predicted probabilities are noticeably better calibrated even when the hard 0.5-threshold predictions come out the same.

5-Fold Cross-Validation Accuracy (Training Set)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9011	0.9560	0.9121	0.9560	0.9560	0.9363
Random Forest	0.9560	1.0000	0.9451	0.9670	0.9890	0.9714

This is where the Random Forest's advantage really shows - it beats the Decision Tree on every single fold, and its worst fold (0.9451) is still better than the Decision Tree's best fold (0.9560) in three out of five cases.

10 Result Visualization

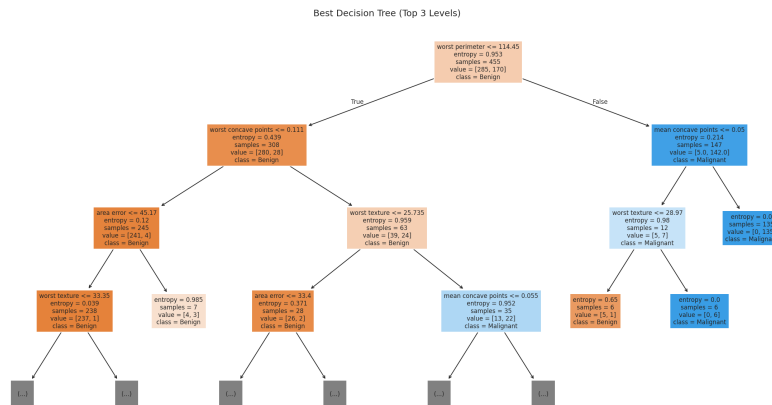


Figure 4: Top three levels of the tuned Decision Tree (`max_depth=5`, entropy criterion).

Inference:

The very first split is on **worst concave points**, which matches what the correlation analysis flagged as one of the two strongest predictors - the tree is essentially confirming what the EDA already suggested. Even after just three levels the tree is doing most of the heavy lifting on class separation, which is a nice illustration of why decision trees can work reasonably well even at fairly shallow depths on a dataset like this one, where the classes are fairly well separated in feature space to begin with.

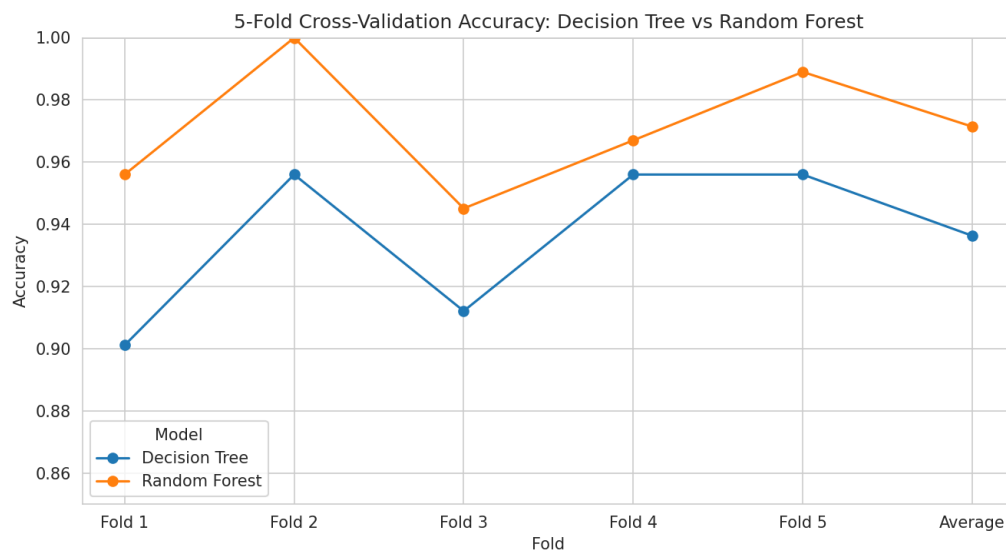


Figure 5: Fold-by-fold 5-fold CV accuracy, Decision Tree vs. Random Forest, on the training set.

Inference:

The Random Forest line sits above the Decision Tree line on every fold without exception, and it's also noticeably less jumpy - the Decision Tree swings between about 0.90 and 0.96 depending on the fold, while the Random Forest stays in a tighter 0.945-1.0 band. That's the variance-reduction effect of bagging in action: averaging over 200 de-correlated trees smooths out the fold-to-fold noise that a single tree is sensitive to.

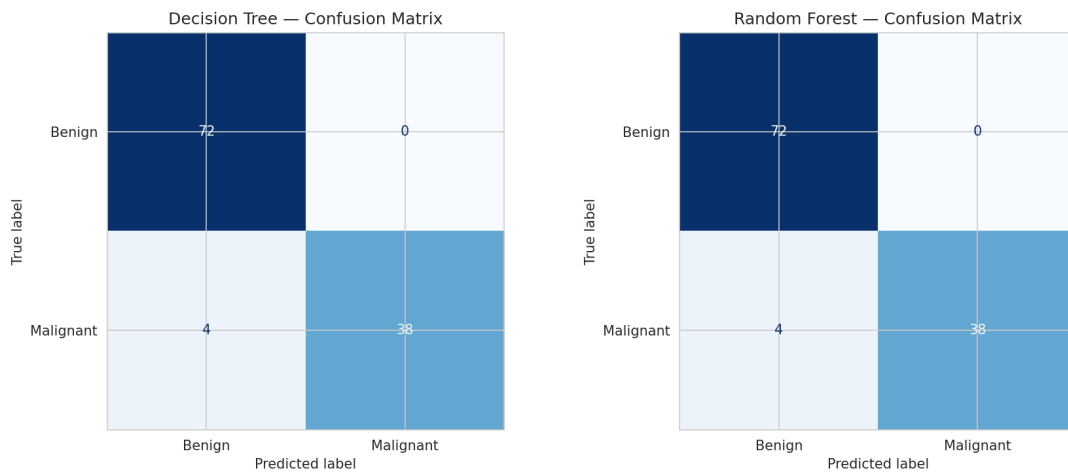


Figure 6: Confusion matrices on the held-out test set for both models.

Inference:

Both models produce the exact same confusion matrix on this test set: all 72 benign cases correctly classified, and 38 of 42 malignant cases caught, with 4 malignant cases missed. Zero false positives (no benign case flagged as malignant) is why precision comes out at a perfect 1.0 for both - but the 4 missed malignant cases are the more clinically important error, since a false negative here means a cancer case that isn't flagged. Neither model fully closes this gap, which is worth flagging honestly rather than glossing over.

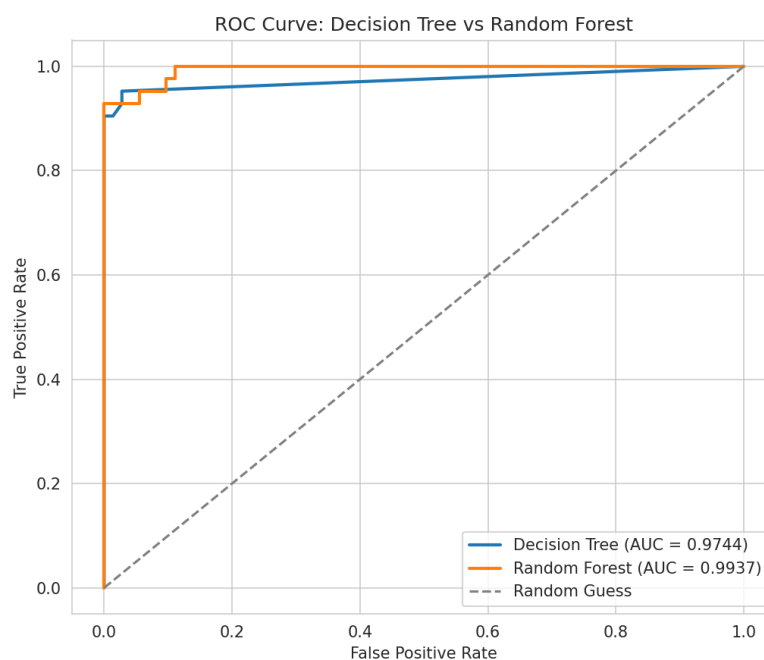


Figure 7: ROC curves for both models on the test set.

Inference:

This is where the Random Forest's edge actually becomes visible: $AUC = 0.9937$ for the Random Forest versus 0.9744 for the Decision Tree. Even though both models make identical hard predictions at the default threshold, the forest's predicted probabilities rank the malignant cases more confidently and consistently - which means that if a different

decision threshold were chosen (e.g. to prioritize catching more malignant cases at the cost of a few more false alarms), the Random Forest would very likely give the better trade-off curve.

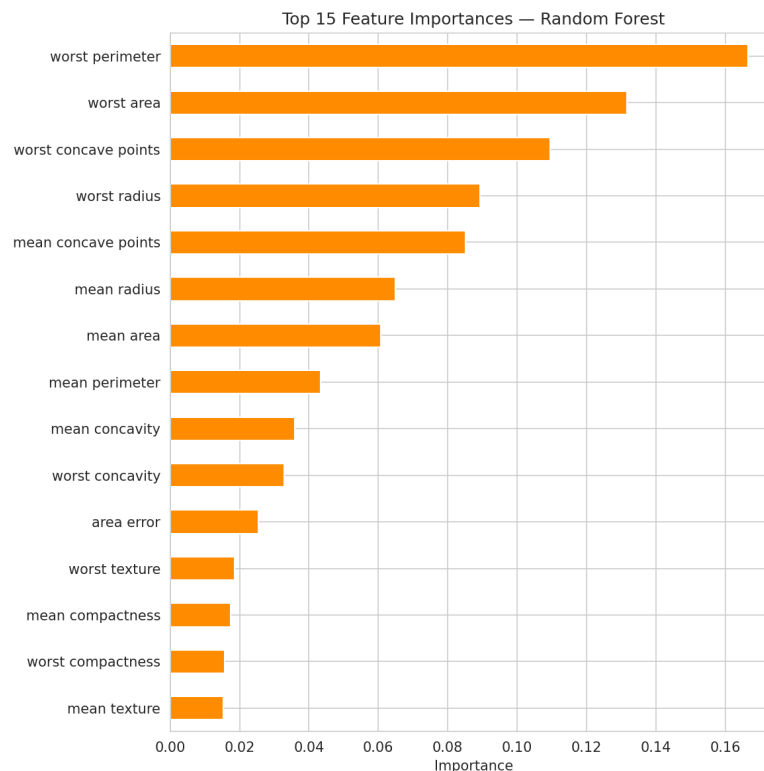


Figure 8: Top 15 feature importances from the Random Forest.

Inference:

worst perimeter, worst area and worst concave points dominate the importance ranking, together accounting for a large share of the total. This is consistent with both the correlation analysis and the root split of the single tree - all three views of the data (correlation, single-tree split, and ensemble importance) are pointing at the same handful of "worst-case" size and shape features as the real drivers of the diagnosis, which is a reassuring consistency check rather than three unrelated coincidences.

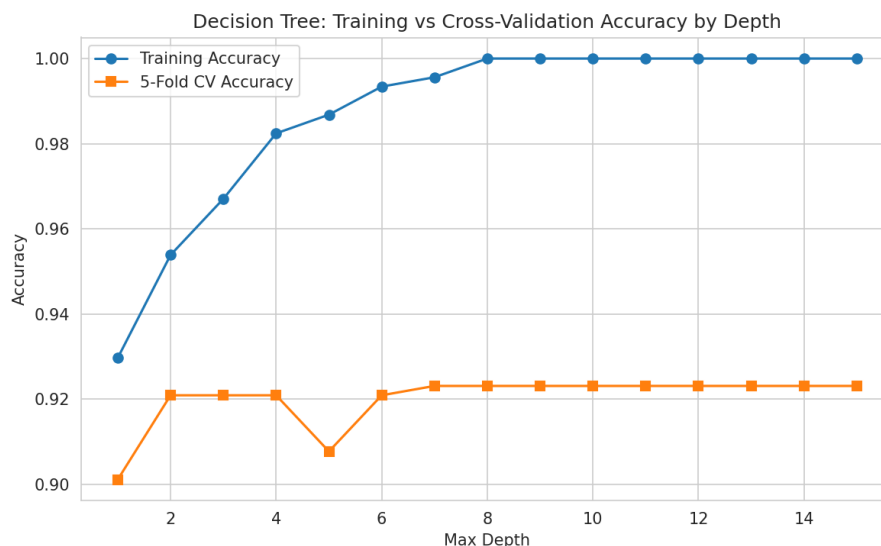


Figure 9: Training vs. 5-fold CV accuracy of a single Decision Tree as `max_depth` increases.

Inference:

This is the clearest overfitting picture in the whole report. Training accuracy climbs steadily and hits a perfect 1.0 by depth 8, while the cross-validation accuracy plateaus around 0.92-0.93 and never gets close. The growing gap between the two curves after about depth 5-6 is the textbook signature of overfitting - the tree starts memorizing noise in the training folds rather than learning anything that generalizes, which is exactly why the grid search picked a comparatively shallow `max_depth=5` rather than letting the tree grow unrestricted.

11 Discussion

A few things stood out while working through this experiment. First, tuning genuinely mattered for the Decision Tree - the untuned baseline tree (default parameters, no depth limit) hit a depth of 8 with 24 leaves and scored 92.98% on the test set, while the depth-5 tuned tree ended up more compact and, on cross-validation, more reliable. Second, on this particular dataset the gap between "the tree does fine" and "the forest does noticeably better" is real but not dramatic - WDBC's classes are already fairly well-separated by a handful of size/shape features, so a single reasonably-tuned tree isn't that far behind. The Random Forest still wins on average CV accuracy (97.14% vs. 93.63%) and ROC-AUC (0.9937 vs. 0.9744), and it's clearly more stable across folds, but on the specific 114-sample test split used here both models land on the same confusion matrix. That's a good reminder that a single test-set snapshot can understate real differences between models - the cross-validation numbers are the more trustworthy comparison. One limitation worth naming: both models miss the same 4 malignant cases, which suggests these are genuinely hard examples (borderline feature values) rather than something that ensembling alone would fix - addressing them further would probably need either more features, class-weighting, or a different decision threshold tuned specifically for recall on the malignant class.

12 Conclusion

Both a Decision Tree and a Random Forest were implemented and tuned via 5-fold cross-validated grid search on the Wisconsin Diagnostic Breast Cancer dataset. The tuned Random Forest (200 trees, `max_depth=5`, `max_features='sqrt'`, no bootstrap) outperformed the tuned Decision Tree (`max_depth=5`, `entropy`) on cross-validation accuracy (97.14% vs. 93.63%) and ROC-AUC (0.9937 vs. 0.9744), while matching it exactly on the held-out test set (96.49% accuracy, 1.0 precision, 0.9048 recall). The depth-vs-accuracy experiment confirmed the expected overfitting behavior of unrestricted Decision Trees, and feature-importance analysis consistently pointed to worst-case perimeter, area and concave-points measurements as the strongest predictors of malignancy across every method used to check it. Overall, the results support the general expectation that ensembling reduces variance and improves generalization over a single tree, even on a dataset where the individual tree was already fairly competitive.

13 References

- [1] scikit-learn developers, “Decision Trees,” scikit-learn documentation: <https://scikit-learn.org/stable/modules/tree.html>
- [2] scikit-learn developers, “Random Forests,” scikit-learn documentation: <https://scikit-learn.org/stable/modules/ensemble.html#forest>
- [3] W. Wolberg, O. Mangasarian, N. Street, and W. Street, “Breast Cancer Wisconsin (Diagnostic) Data Set,” UCI Machine Learning Repository: <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>